

Hanzhang Yuan

3060 W Olympic Blvd, Los Angeles, CA, 90006 (909)551-8663

yuanhanzhang21@gmail.com

SUMMARY

I am a UCLA Applied Statistics and Data Science graduate student with strong training in mathematics and computer science, experienced in building end-to-end machine learning systems such as NLP pipelines, RAG systems, and reinforcement learning applications.

EDUCATION

MS, **Applied Statistics and Data Science** (GPA: 3.93/4.0) *University of California, Los Angeles*

Sep 2025 - Dec 2026

BS, **Mathematics and Minor Computer Science** (GPA: 3.45/4.0) *University of Illinois at Urbana-Champaign* Aug 2021 - May 2025

WORKING EXPERIENCE

ETH Tech

Newark, CA

Intern Project : Amazon Food Review Sentiment Analysis

Jun 2025 - Aug 2025

- Built an end-to-end **sentiment analysis pipeline** on ~500K Amazon Fine Food Reviews using Python, Pandas, NLTK, and Scikit-learn to extract insights from large-scale customer feedback.
- Performed comprehensive text preprocessing (tokenization, lowercasing, stopwords removal, lemmatization) and engineered **TF-IDF** features with unigrams and bigrams to enhance model performance.
- Trained and compared multiple classifiers (**Logistic Regression**, Linear SVM, Random Forest), selecting the optimal model based on F1-score, Precision, and Recall, achieving 92% accuracy and a 0.89 F1-score on the test set.
- Designed **interactive dashboards** using Power BI and Matplotlib to visualize sentiment trends, review distributions, and misclassification patterns, presenting insights to crossfunctional stakeholders to inform next-generation product design.

NextTier

Sacramento, CA

End-to-End Retrieval-Augmented Generation (RAG) System for Large-Scale Web Knowledge

Dec 2025 - Present

- Built an automated web crawling pipeline using BeautifulSoup to ingest full content from the Shopify official blog, implementing daily incremental tracking, checkpointing, and **loop-prevention** logic to ensure stable, repeatable data collection without redundant or infinite crawling.
- Designed a Pinecone vector database **metadata schema** to support efficient semantic retrieval and downstream reranking, enabling structured storage of embeddings, document context, and filtering attributes for scalable retrieval workflows.
- Designed and implemented an end-to-end semantic retrieval loop integrating chunking, retrieval, and reranking logic, and closed the loop with an automated **evaluation system** to identify optimal chunk sizes, Top-N retrieval depth, and reranking configurations.
- Built interactive user interfaces using Gradio and Streamlit, and packaged the RAG pipeline into a functional **application**, enabling real-time querying, result inspection, and rapid experimentation.
- Leveraged Ragas to generate standardized question - answer evaluation pairs and quantify retrieval and generation quality, enabling data-driven **iteration** across chunking, retrieval, and reranking components.

RESEARCH EXPERIENCE

UCLA — Reinforcement Learning for Glucose Regulation (TD3)

Sep 2025 - Present

Research Project under Prof. Yuhua Zhu

Research Description:

Designed and implemented a reinforcement learning system to regulate insulin micro-dosing by optimizing long-horizon glucose trajectories. The objective is to use RL algorithms (TD3, which is classic RL Algorithm) to learn an optimal dosing policy from CGM-insulin historical datasets.

- Implemented RL algorithms (TD3) and compared them against traditional regression approaches, highlighting RL's ability to learn individualized insulin-adjustment policies directly from CGM trajectories.
- Designed a clinically meaningful reward function that penalizes hypo/hyperglycemia, rapid glucose swings, and unsafe dosing changes while rewarding stable, in-range glucose trajectories. This shaped the policy toward safer and more physiologically realistic insulin recommendations.
- Selected the final model by evaluating candidate policies within a physiologically grounded glucose - insulin simulator, enabling safe comparison of control performance, hypoglycemia risk, and dosing behavior without real-world patient experimentation.

SELECTED PROJECTS

Junior's Project : Risk Modeling for International Trade Credit

Project description:

Built a risk scoring framework for international trade credit, integrating transaction records, firm profiles, and compliance data through a normalized SQL schema. Transforming raw transaction and firm data into model-ready features, with automated checks for

missing, invalid, and inconsistent values. Trained and evaluated a logistic regression model to score Letters of Credit applications and support credit decision-making.

- Designed a **normalized SQL schema** and developed a feature pipeline to integrate transaction records, firm profiles, and compliance history.
- Engineered mixed-type features and implemented **automated data quality validation** in Python to ensure feature integrity.
- Built and evaluated a **logistic regression model** to score Letters of Credit applications, optimizing thresholds and achieving high accuracy on holdout data.

Senior's Project : Automated Financial Statement Consistency Checker

Project Description :

Built a Python-based data science pipeline to preprocess, reconcile, and analyze multiperiod financial statement data, combining statistical modeling and anomaly detection to quantify reporting reliability for downstream credit and bond eligibility analysis.

- Developed a **Python-based verification pipeline** using Pandas and NumPy to cross-check annual financial statements against monthly and quarterly reports, ensuring reporting completeness and data integrity.
- Built a **consistency-scoring model** leveraging logistic regression and z-score anomaly detection to quantify reporting reliability and support bond issuance eligibility assessments.
- Conducted **statistical validation** using hypothesis testing (t-tests, chi-square) and implemented an automated reconciliation system to align data with credit risk evaluation standards, enhancing audit readiness.
- Produced structured, **audit-ready reports** summarizing mismatches and reliability scores, accelerating decision-making workflows by 35% and improving bond approval efficiency

SKILL LIST

- **Programming Languages : C++; Python(PyTorch, NumPy, Pandas); Java; SQL; R**
- **Data Science** : Implemented **linear algebra** concepts and **algorithms** using **Python**; Familiar with **probability theory** and its applications.
- **Computer Science** : Proficient in **database design, querying, and optimization**; Experienced in **data structures** and their applications
- **Machine Learning: Reinforcement Learning, Supervised Learning (Linear/Logistic Regression, Decision Trees)**
- **Related Courses** : Statistics and Probability I; Data Structures; Elementary Real Analysis; Abstract Linear Algebra; Numerical Methods; Database Systems; Probability Theory